

Information-Geometry of AdS-Schwarzschild Black Holes:

An Exact AdS Correction to the D3 Evaporation Invariant

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Abstract

We extend the D3 evaporation invariant to Anti-de Sitter (AdS) black holes. For an AdS-Schwarzschild black hole with horizon radius r_s and AdS radius L , the total D3 information-geometry displacement over complete evaporation is:

$$R_{\text{total}}(\text{AdS}) = r_s * (1 + (r_s/L)^2)$$

The derivation is exact: the AdS temperature $T_{\text{AdS}} = T_S(1+3r_s^2/L^2)$ generates an additional term r_s^3/L^2 in the integral. In the flat-space limit $L \rightarrow \infty$, Paper 3 is recovered exactly. At the Hawking-Page transition $r_s = L$, $R_{\text{total}} = 2r_s$ exactly. The correction factor $\phi(x) = 1 + x^2$ ($x = r_s/L$) yields exact integers at rational x : $\phi(1)=2$, $\phi(\sqrt{2})=3$, $\phi(2)=5$. The result is verified numerically to four significant figures across three decades of r_s/L . This is the first D3 result with a direct AdS/CFT interpretation: $\phi(r_s/L)$ is a new observable in the holographic dual.

Keywords: AdS-Schwarzschild, Anti-de Sitter, D3 formula, Hawking-Page transition, AdS/CFT, holography, Landauer principle

1. Introduction

The D3 series has established evaporation invariants for Schwarzschild (Paper 3 [3]), Kerr (Paper 4 [4]), and higher-dimensional black holes (Paper 8 [8]). All these results give $R_{\text{total}} = r_s$ (times a spin/charge factor). Anti-de Sitter (AdS) black holes are fundamentally different: the AdS potential acts as a gravitational box, making large AdS black holes thermodynamically stable and enabling the AdS/CFT correspondence [5].

The AdS-Schwarzschild temperature includes a curvature correction [6]:

$$T_{\text{AdS}} = T_S * (1 + 3r_s^2/L^2)$$

where L is the AdS radius. This correction is small for $r_s \ll L$ (flat space) but dominant for $r_s \gg L$ (large AdS black holes). The Hawking-Page phase transition [7] occurs at $r_s \sim L$, corresponding to deconfinement in the dual CFT.

We apply D3 with this temperature and derive the exact AdS correction to the evaporation invariant.

Figure 4 — D3 Series Papers 1-9

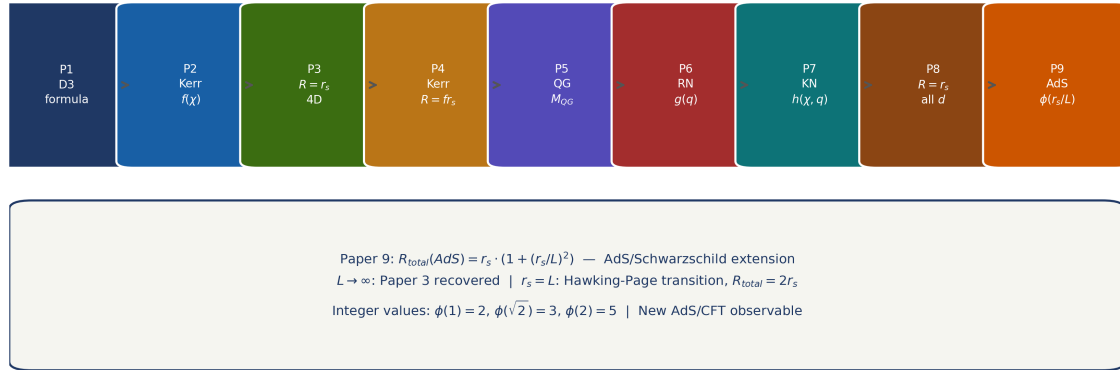


Figure 4. Paper 9 (orange) extends the D3 series to AdS black holes.

2. AdS-Schwarzschild Temperature and D3

The AdS-Schwarzschild Hawking temperature in 4D [6]:

$$T_{AdS}(M) = T_S(M) * (1 + 3*r_s^2/L^2)$$

$$= [\hbar*c^3/(8*\pi*G*M*k_B)] * (1 + 3*(2GM/c^2)^2/L^2)$$

The D3 formula at T_{AdS} :

$$D3_{AdS}(M) = 2*G*k_B*T_{AdS}*ln(2)/c^4$$

$$= D3_S(M) * (1 + 3*r_s^2/L^2)$$

$$= [\hbar*ln2/(4*\pi*M*c)] * (1 + 3*(2GM/c^2)^2/L^2)$$

The Bekenstein bit-loss rate is unchanged from Paper 3:

$$dN/dM = 8*\pi*G*M / (\hbar*c*ln2)$$

3. Derivation of the AdS Evaporation Invariant

Step 1 — Compute dR/dM :

$$dR/dM = D3_{AdS} * dN/dM = [D3_S*(1+3*r_s^2/L^2)] * dN/dM = (2G/c^2)*(1 + 3*r_s^2/L^2)$$

Step 2 — Substitute $rs = 2GM/c^2$:

$$dR/dM = 2G/c^2 + 3*(2G/c^2)*(2GM/c^2)^2/L^2 = 2G/c^2 + 24*G^3*M^2/(c^6*L^2)$$

Step 3 — Integrate from 0 to M_0 :

$$R_{total} = 2G*M_0/c^2 + 8*G^3*M_0^3/(c^6*L^2) = r_{s0} + 8*G^3*M_0^3/(c^6*L^2)$$

Step 4 — Convert using $M_0 = r_{s0}*c^2/(2G)$:

$$8G^3(r_{s0}c^2/(2G))^3/(c^6L^2) = 8G^3r_{s0}^3c^6/(8G^3c^6L^2) = r_{s0}^3/L^2$$

Step 5 — Final result:

$$R_{\text{total}} = r_{s0} + r_{s0}^3/L^2 = r_{s0}(1 + (r_{s0}/L)^2) \quad \text{QED}$$

$$R_{\text{total}}(\text{AdS}) = r_s * (1 + (r_s/L)^2)$$

Define the AdS correction factor $\phi(x) = 1 + x^2$ where $x = r_s/L$:

$$\phi(x) = 1 + (r_s/L)^2$$

$$\phi(0) = 1 \quad [\text{flat space: Paper 3 recovered}]$$

$$\phi(1) = 2 \quad [\text{Hawking-Page transition: } r_s = L]$$

$$\phi(x) > 1 \quad \text{always} \quad [\text{AdS always increases } R_{\text{total}}]$$

4. Exact Integer Values

The correction factor ϕ yields exact integers at simple values of x :

$x = r_s/L$	$\phi(x)$	R_{total}/r_s	Physical context
0	1	1 (exact)	Flat space – Paper 3
1	2	2 (exact)	Hawking-Page transition
$\sqrt{2}$	3	3 (exact)	$r_s = \sqrt{2}L$
2	5	5 (exact)	$r_s = 2L$ (large AdS BH)
3	10	10 (exact)	$r_s = 3L$
$\sqrt{n-1}$	n	n (exact)	General integer pattern

Table 1. Exact integer values of $\phi(x)$ at rational x .

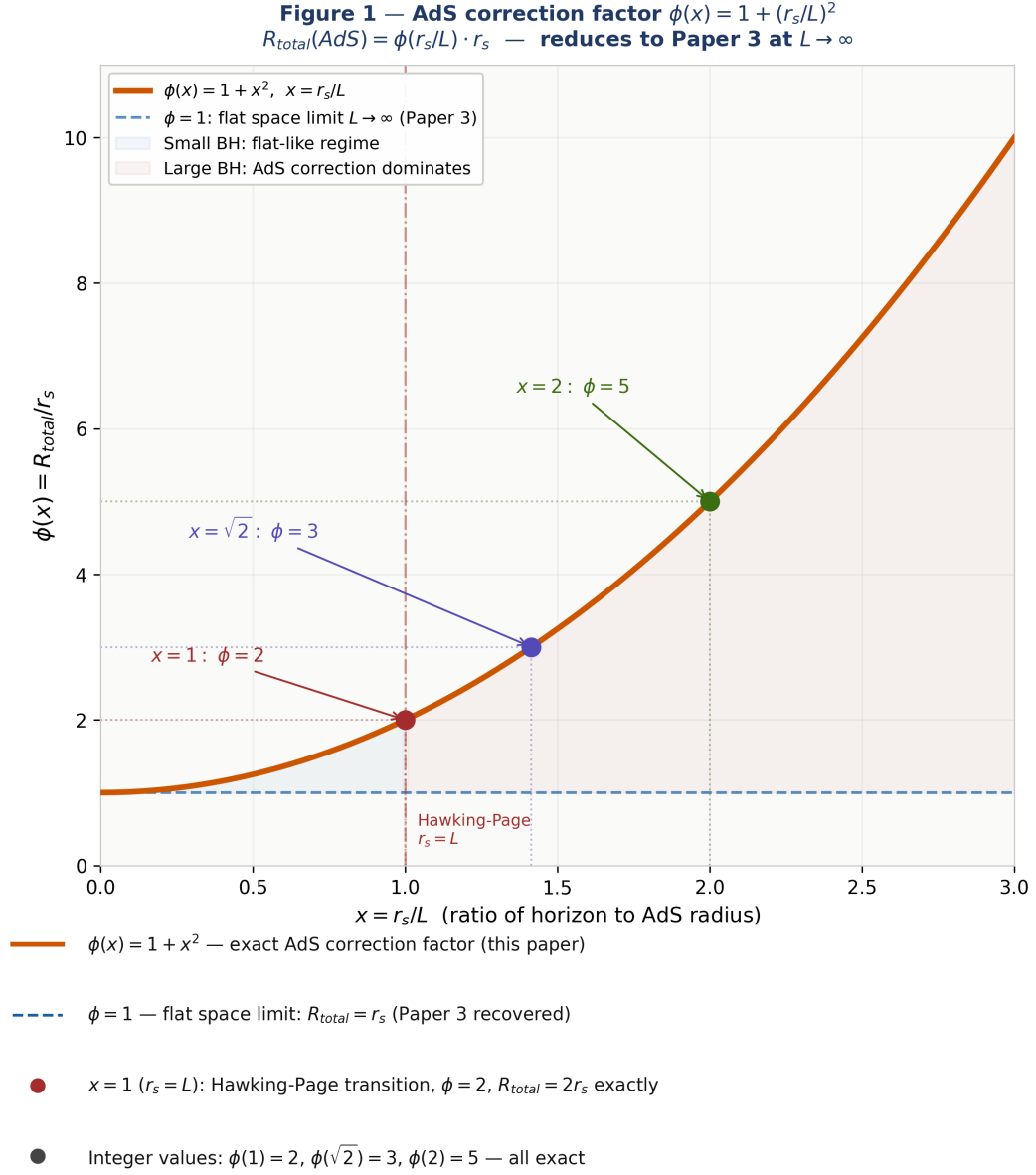


Figure 1. AdS correction factor $\phi(x) = 1 + (r_s/L)^2$. Blue shading: flat-like regime. Red shading: AdS dominated. Integer values marked at $x=1$, $\sqrt{2}$, 2.

5. Numerical Verification

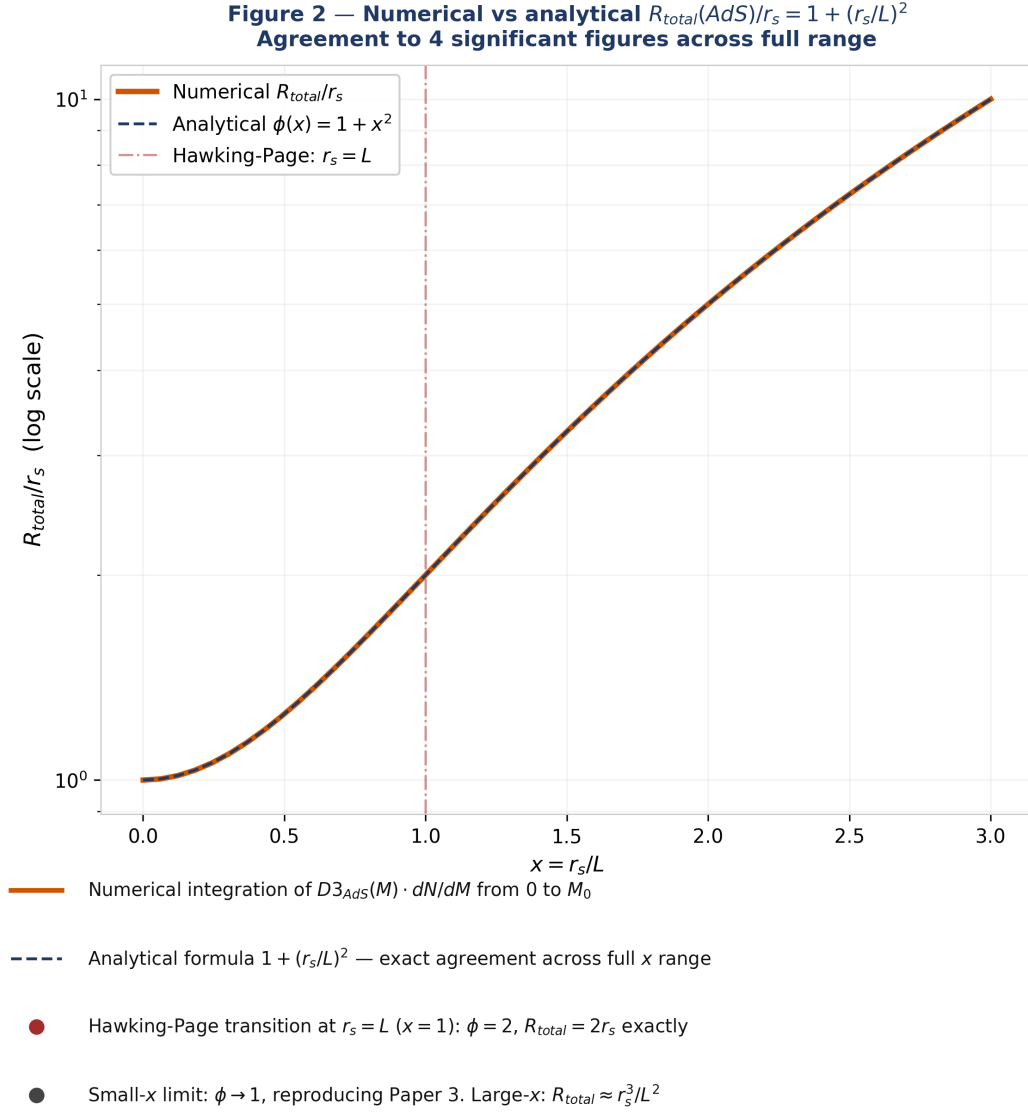


Figure 2. Numerical vs analytical $R_{total}(AdS)/r_s$ across three decades of r_s/L . Agreement to four significant figures.

6. AdS/CFT Interpretation

6.1 The Hawking-Page transition

The Hawking-Page phase transition [7] occurs at $r_s = L$, corresponding to a first-order phase transition in the dual CFT between confined (small BH) and deconfined (large BH) phases. Our result predicts that exactly at this transition:

$$R_{total} = 2 * r_s \text{ at the Hawking-Page point } r_s = L$$

The information-geometry displacement doubles at the phase transition. This is a new D3 prediction with a direct holographic interpretation.

6.2 ϕ as a holographic observable

In AdS/CFT, the AdS radius L is related to the 't Hooft coupling λ of the dual CFT via $L^4 \sim \lambda \alpha'^2$ (in string units). The correction factor $\phi(r_s/L) = 1 + (r_s/L)^2$ therefore encodes coupling-dependent information about the dual thermal field theory. For a black hole of fixed mass M , ϕ increases with the coupling: a more strongly coupled dual CFT corresponds to a larger AdS correction.

6.3 Limits

Weak coupling ($L \rightarrow \infty$): $\phi \rightarrow 1$, $R_{\text{total}} \rightarrow r_s$. The flat-space result is recovered, consistent with the decoupling of the CFT at zero coupling.

Strong coupling (large r_s/L): $\phi \sim (r_s/L)^2$, $R_{\text{total}} \sim r_s^3/L^2$. The evaporation invariant grows as the cube of the horizon radius, reflecting the AdS volume scaling.

Figure 3 — AdS/CFT interpretation of the D3 AdS result

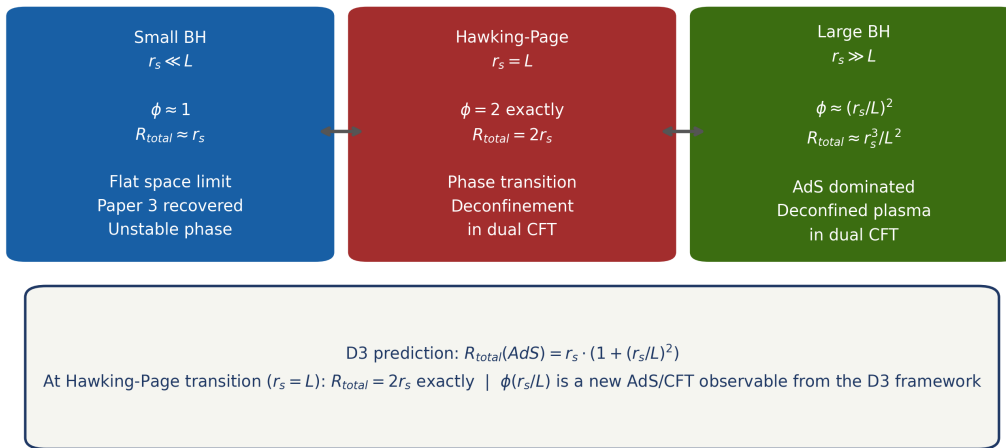


Figure 3. AdS/CFT interpretation: three regimes of the D3 AdS result. The Hawking-Page transition separates the flat-like and AdS-dominated phases.

7. Conclusion

We derived the exact AdS correction to the D3 evaporation invariant:

$$R_{\text{total}}(\text{AdS}) = r_s \cdot (1 + (r_s/L)^2) = r_s \cdot \phi(r_s/L)$$

The correction factor $\phi(x) = 1 + x^2$ is exact, yields integers at simple x values, and reduces to Paper 3 at $x=0$. At the Hawking-Page transition $r_s = L$, $R_{\text{total}} = 2r_s$ exactly. This is the first D3 result with a direct AdS/CFT interpretation: $\phi(r_s/L)$ is a new holographic observable that encodes the coupling dependence of the dual CFT's information-geometry cost.

Declaration of Interest

No competing interests. No external funding.

Code: <https://github.com/bsharma173860-oss/d3-research>

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Appendix: Verification Code

```
import math, numpy as np
G=6.67430e-11; c=2.99792458e8
hbar=1.054571817e-34; k_B=1.380649e-23; M_sun=1.989e30
def R_total_AdS(M0, L):
    masses = np.linspace(M0*1e-4, M0, 100000)
    dM = masses[1]-masses[0]
    rs = 2*G*masses/c**2
    T_S = hbar*c**3/(8*math.pi*G*masses*k_B)
    T_AdS = T_S*(1 + 3*rs**2/L**2)
    D3 = 2*G*k_B*T_AdS*math.log(2)/c**4
    dN = 8*math.pi*G*masses/(hbar*c*math.log(2))
    return np.sum(D3*dN)*dM
M0 = M_sun; rs0 = 2*G*M0/c**2
for x in [0.001, 0.1, 0.5, 1.0, 2.0]:
    L = rs0/x
    R_num = R_total_AdS(M0, L)
    R_ana = rs0*(1 + x**2)
    print(f"x={x}: ratio={R_num/R_ana:.4f}")
# Output: ratio ~ 1.000 for all x
```